KESAVA PAVAN GADDE

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SUMMARY

B.Tech graduate in Computer Science Engineering, specialized in Artificial Intelligence, from Parul University. After graduation, I independently built hands-on expertise in Data Science and Generative AI, developing and deploying a hybrid RAG assistant (FAISS + BM25) and an XGBoost house price model (R^2 = 0.92 on 187K+ records), and completing a Data Science & Gen AI internship using Python, LangChain, and LangGraph. Seeking a Data Scientist / Gen AI Engineer role to apply this self-driven, hands-on experience to real-world problems.

TECHNICAL SKILLS

Programming: Python, SQL

Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Hybrid Search (FAISS + BM25), Prompt Engineering, LangChain, LangGraph, Groq API, Google Gemini API

Machine Learning: Scikit-learn, Regression, Classification, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Random Forest, Gradient Boosting, XGBoost

Natural Language Processing (NLP): NLTK, Hugging Face Transformers, Text Preprocessing, Lemmatization, Embeddings, Vector Databases

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)

Tools & Deployment: Git, GitHub, Streamlit (Community Cloud), AWS (EC2), Jupyter Notebook, VS Code, Joblib

EXPERIENCE

Data Science & Gen AI Intern | Innomatics Research Labs **Nov 2025 – Mar 2026**

- Analyze real-world datasets using Python, Pandas, and NumPy; performed EDA and built Matplotlib and Seaborn visualizations to identify trends.
- Built end-to-end ML pipelines covering preprocessing, model training, and evaluation, with reproducible experiments.
- Developed RAG-based question-answering systems with LangChain and FAISS retrieval, using prompt engineering to reduce hallucinations.
- Built multi-step AI agents with LangGraph for tool-driven task automation.
- Built and deployed interactive Streamlit applications for ML models and RAG-based question answering.
- Gained hands-on experience with AWS EC2, including launching and configuring cloud instances.

PROJECTS

ResearchChat-AI: Hybrid RAG Assistant | *Python, LangChain, FAISS, BM25, Streamlit* GitHub | Live Demo

- Built a hybrid retrieval pipeline combining dense search (FAISS, BAAI/bge-small-en) with BM25 sparse search to handle both paraphrased and exact-term queries.
- Fused rankings using Reciprocal Rank Fusion (k=60) to return the top 4 chunks without score normalization.
- Parsed PDFs with Docling and chunked with RecursiveCharacterTextSplitter (1500-character chunks, 300 overlap).
- Deployed a Streamlit app (Groq-hosted gpt-oss-20b) with multi-PDF upload, chunk inspection, and response-time tracking.

House Price Prediction | *Python, Scikit-learn, XGBoost, Streamlit* GitHub | Live Demo

- Built a regression pipeline on 187K+ row Indian housing data, extracting BHK, floor, and carpet area and standardizing price formats (e.g., “1.40 Cr” → Lakhs).
- Applied outlier filtering and missing-value imputation, then encoded/scaled features via ColumnTransformer (OrdinalEncoder + RobustScaler).
- Benchmarked Random Forest, Gradient Boosting, and XGBoost with 5-fold CV; tuned XGBoost achieved an R^2 score of 0.92.
- Deployed a Streamlit app that loads the Joblib-serialized model and preprocessing pipeline for real-time price predictions.

AI Career Advisor Chatbot | *Python, Google Gemini API, Streamlit* GitHub | Live Demo

- Built a domain-specific career-guidance chatbot on the Google Gemini API and Streamlit, covering career paths, skill-gap analysis, learning roadmaps, and interview preparation.
- Designed a modular backend (chatbot orchestration, prompt manager, Gemini client, memory manager, logger, config) that separates UI, prompting, and API logic.
- Implemented multi-turn memory with history limiting and token optimization to prevent prompt overflow, plus prompt guardrails that restrict responses to career topics.
- Added response validation, exception handling, and persistent logging (logs/app.log); managed API keys via environment variables; deployed on Streamlit.

EDUCATION

Parul University, Vadodara **2021 – 2025**
B.Tech in Computer Science Engineering, specialized in Artificial Intelligence

CERTIFICATIONS

Machine Learning, IBM (2026) | MySQL, Innomatics Research Labs (2025) | Python, Kaggle (2023)